

Lab Report

ACC

Participants:

1. Muhammad Arhum Siddiqui mus8857
2. Waleed wam1633
3. Zarrar Feroze Khan zuk1232
4. Muhammad Ahmed Khan muk9044
5. Aftab Nabi Shaikh Afs7221

Group-Number: 5

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<i>Experiment</i>	<i>Date of Experiment</i>	<i>Date of Submission</i>
ACC	20th April 2026	26th April 2026
CISS		
HIL	9th April 2026	14th April 2026
RADAR		
SENSOR&ACTUATOR		
STEADY CIRCLE	16th April 2026	24th April 2026

1 Experiment / Execution

The following section presents the results of all test runs performed in the CarMaker driving simulator. A total of five route scenarios were tested, each with all three available radar sensors (RadarL_1, RadarL_2, and RadarL_3). The variable plotted on the Y-axis is the longitudinal distance to the target vehicle, measured via the sensor signal "Sensor.Object.RadarL.relvTgt.Near.Pnt.ds.x" [m], and the X-axis represents the road position "Vhcl.sRoad" [m]. All test runs completed successfully, as confirmed by the green result indicators in the CarMaker Test Manager.

1.1 Test Manager Overview

The screenshots below show the CarMaker Test Manager after the completion of all test runs. Every test scenario across all sensor configurations received a green result indicator, confirming that each simulation ran to completion without critical errors. The three sensors (RadarL_1, RadarL_2, RadarL_3) were assigned the following parameter sets:

Sensor	FoV_h (Azimuth) [°]	FoV_v (Elevation) [°]	Rmax (Range) [m]	Label
RadarL_1	30	3	200	Sensor 1
RadarL_2	90	15	120	Sensor 2
RadarL_3	170	30	80	Sensor 3

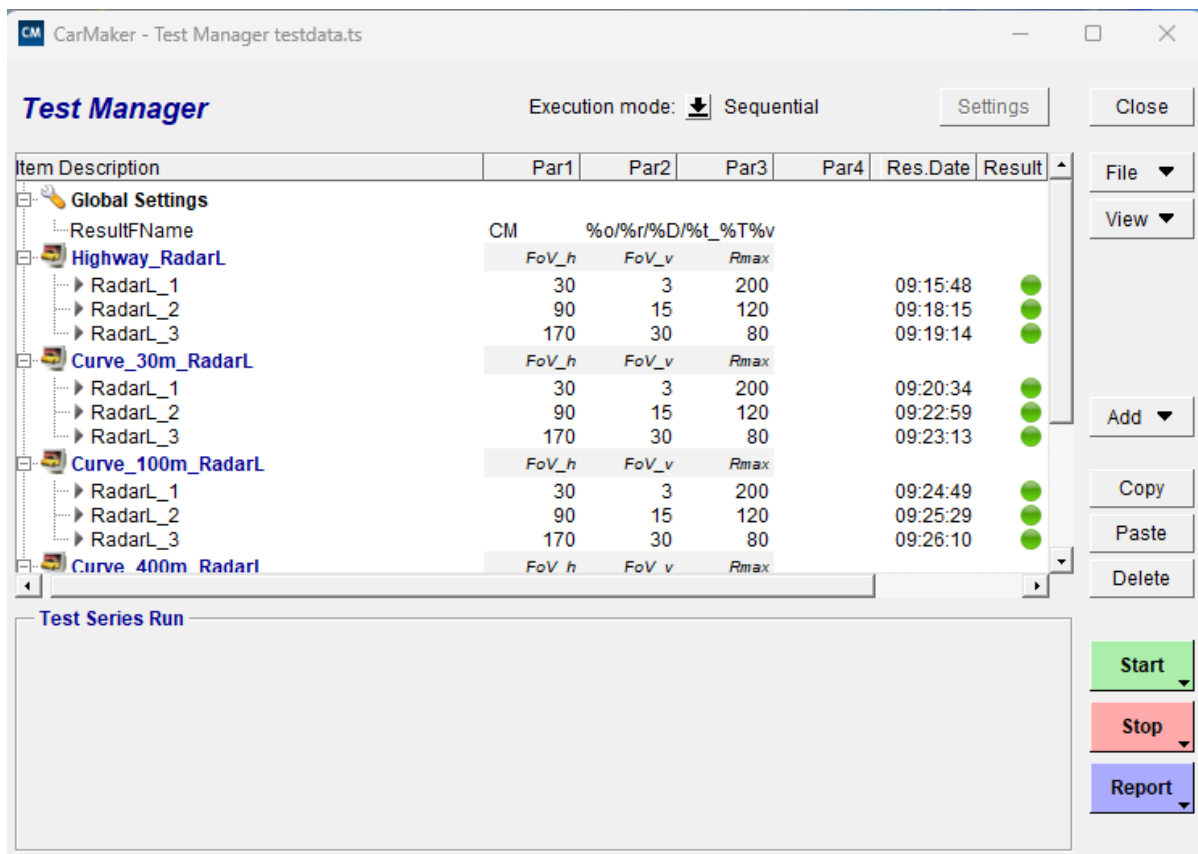


Figure 1: CarMaker Test Manager – Test runs for Highway, Curve 30m, and Curve 100m scenarios (all green)

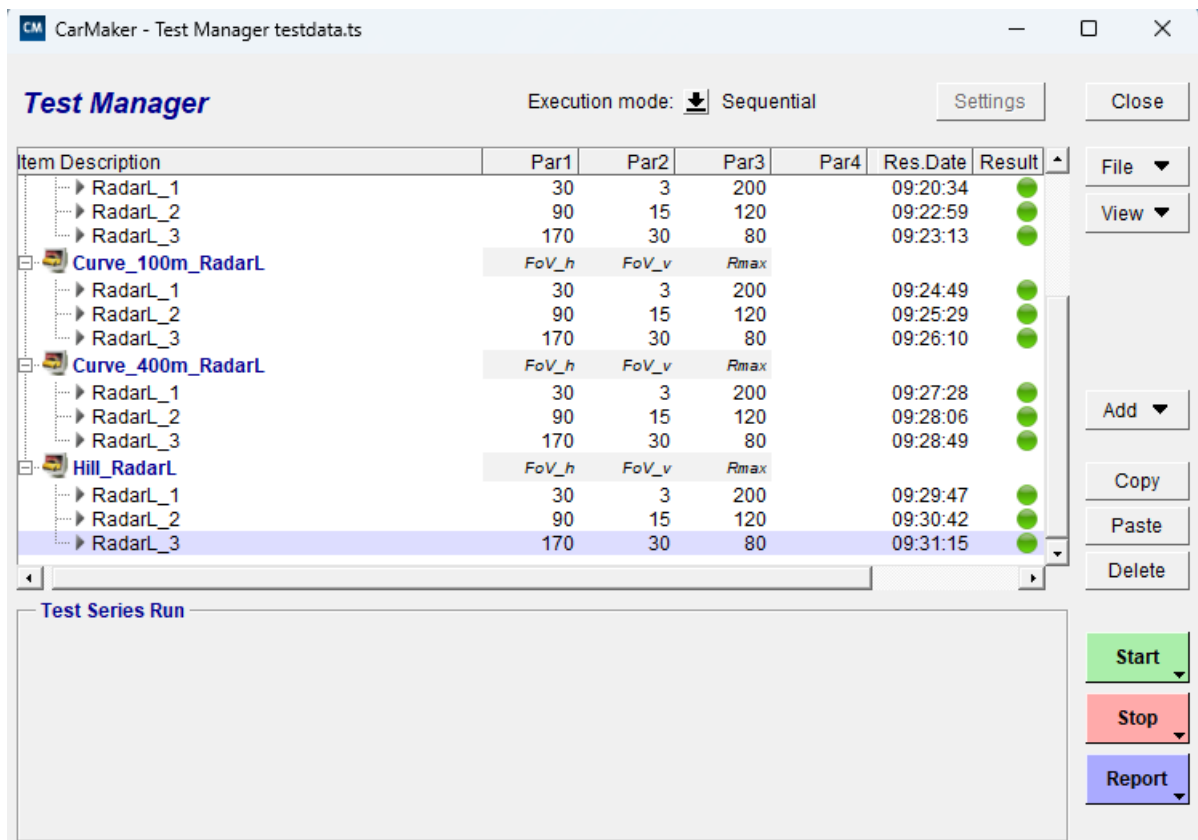


Figure 2: CarMaker Test Manager – Test runs for Curve 400m and Hill scenarios (all green)

Figures 1 and 2 together display the complete Test Manager view. All five test groups — Highway_RadarL, Curve_30m_RadarL, Curve_100m_RadarL, Curve_400m_RadarL, and Hill_RadarL — were executed sequentially with each of the three sensor configurations. The timestamps confirm sequential execution. A green dot in the Result column indicates the simulation ran without fatal errors; it does not indicate that the ACC system met the distance-keeping requirement.

1.2 Highway Scenario

Scenario parameters: Ergo_Car_v0 = 200 km/h (ego vehicle set speed), Object_Car_v = 130 km/h (vehicle ahead). The ego vehicle approaches the slower object car from behind on a straight highway and the ACC must decelerate and maintain the required time-gap distance of $d_x = v_x \cdot t_l = (130/3.6) \cdot 1.8 \approx 65$ m. The X-axis shows road distance traveled by the ego vehicle [m]; the Y-axis shows the longitudinal sensor distance to the target vehicle [m].

1.2.1 Highway – Sensor 1 (RadarL_1: Range 200 m, Azimuth 30°, Elevation 3°)

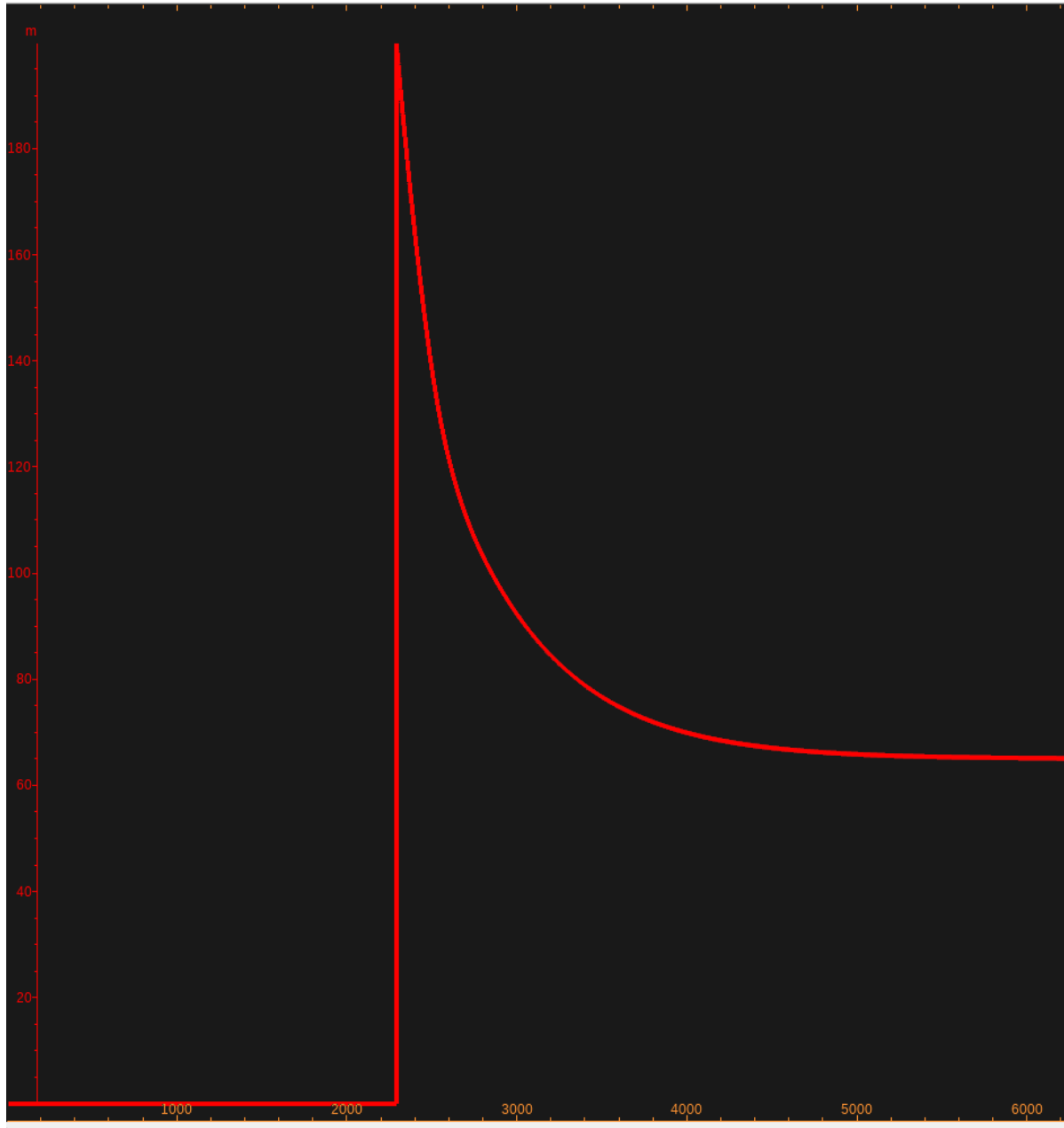


Figure 3: Highway scenario – Sensor 1 (RadarL_1) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

The graph shows a characteristic approach-and-follow pattern. The signal is flat at zero until approximately $sRoad = 2200$ m, at which point the ego vehicle first detects the object car — the measured distance jumps sharply to approximately 190 m, indicating detection at near the sensor's maximum range of 200 m. This confirms that Sensor 1 achieves early, long-range detection of the target. Following detection, the measured distance decays rapidly as the ACC decelerates the ego vehicle from 200 km/h to match the 130 km/h object car speed. The curve then flattens and stabilizes at approximately 65 m, which corresponds precisely to the required time-gap distance $d_x = 130/3.6 \times 1.8 \approx 65$ m. This confirms that Sensor 1 successfully enables the ACC system to fulfill the highway approach requirement.

1.2.2 Highway – Sensor 2 (RadarL_2: Range 120 m, Azimuth 90°, Elevation 15°)

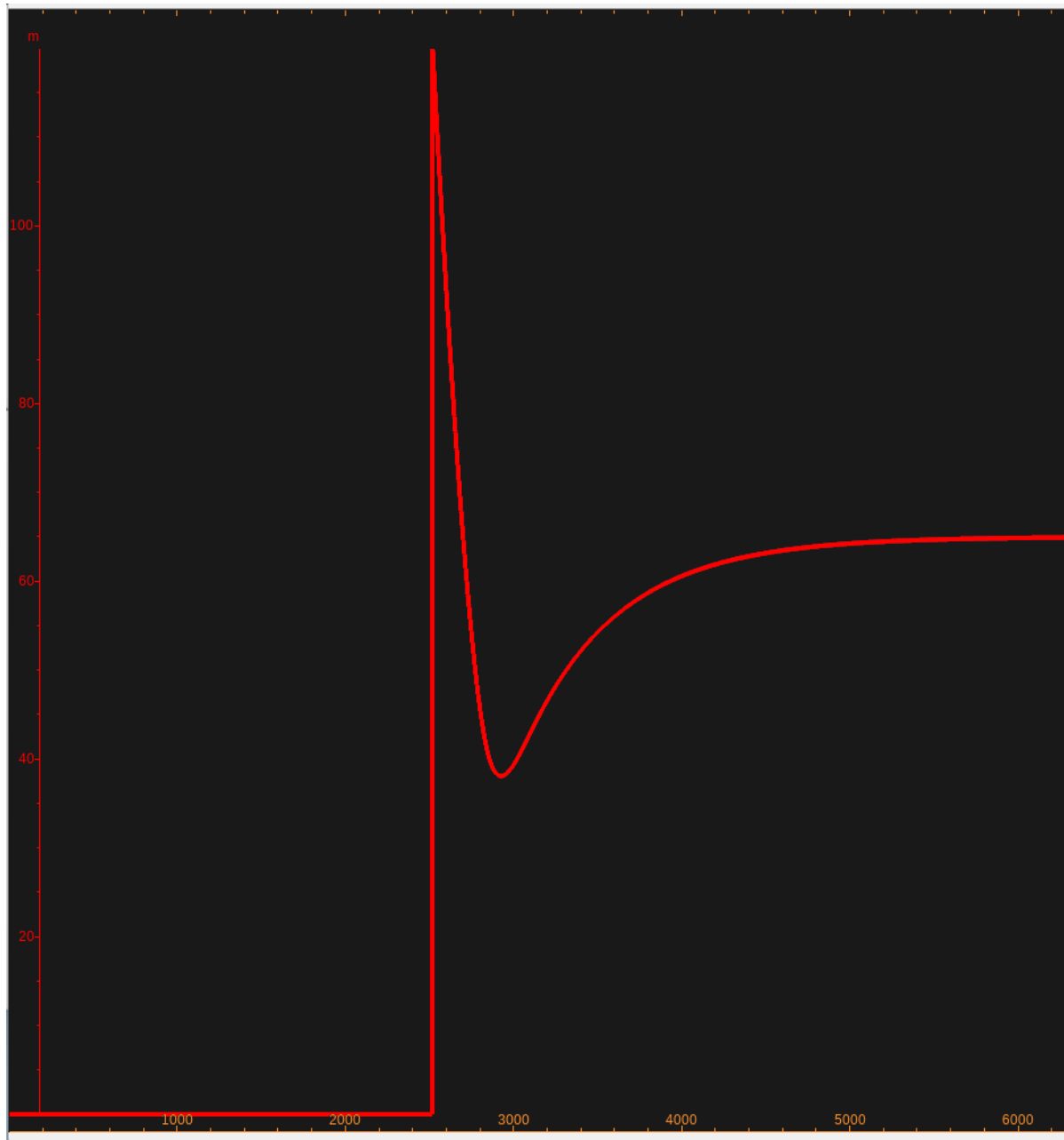


Figure 4: Highway scenario – Sensor 2 (RadarL_2) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

The graph again begins flat at zero until around $sRoad = 2200$ m, after which detection occurs. However, the initial detected distance peaks at approximately 115 m — significantly lower than Sensor 1 — consistent with Sensor 2's reduced maximum range of 120 m. After detection, the distance first decreases sharply to a minimum of approximately 38 m before recovering and stabilizing. This undershoot indicates that the ACC did not detect the vehicle early enough and overshot the target distance, requiring a more aggressive deceleration. The system eventually recovers and settles at approximately 65 m. While the ACC eventually maintains the correct distance, the late detection leads to a noticeable undershoot, meaning the system barely avoids getting dangerously close to the lead vehicle. The highway requirement is marginally met, but with a lower safety margin than Sensor 1.

1.2.3 Highway – Sensor 3 (RadarL_3: Range 80 m, Azimuth 170°, Elevation 30°)

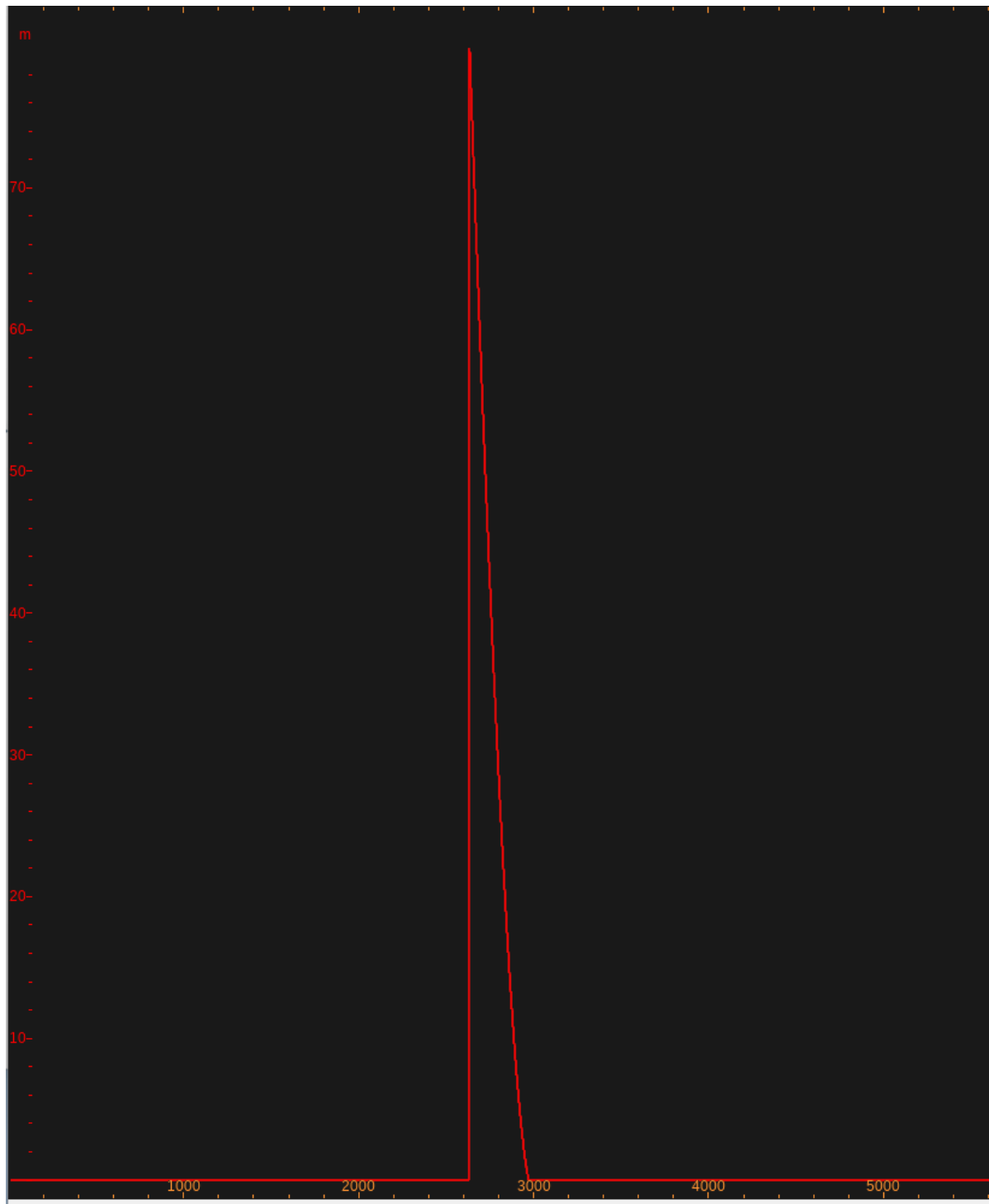


Figure 5: Highway scenario – Sensor 3 (RadarL_3) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

The graph shows that Sensor 3 detects the object car at around $sRoad = 2250$ m, with a peak measured distance of approximately 75 m — close to the sensor's maximum range of 80 m. After detection, the measured distance drops almost vertically and reaches zero near $sRoad = 2900$ m. This near-zero (or collision-level) distance means the ego vehicle's ACC system was unable to react in time to decelerate from 200 km/h to 130 km/h within the available detection distance. The

very short range of 80 m gives the ACC insufficient time and distance to brake comfortably at the allowed $\pm 2.5 \text{ m/s}^2$ deceleration limit. The highway scenario requirement is therefore NOT fulfilled by Sensor 3.

1.3 Curve 30 m Radius Scenario

Scenario parameters: Ergo_Car_v0 = 60 km/h, Object_Car_v = 30 km/h. Required time-gap distance: $d_x = 30/3.6 \times 1.8 \approx 15 \text{ m}$. This is a very tight circle with a small radius, requiring the sensor to have a sufficiently wide azimuth (horizontal field of view) to keep the lead vehicle in view as it curves away.

1.3.1 Curve 30m – Sensor 1 (RadarL_1: Range 200 m, Azimuth 30°, Elevation 3°)

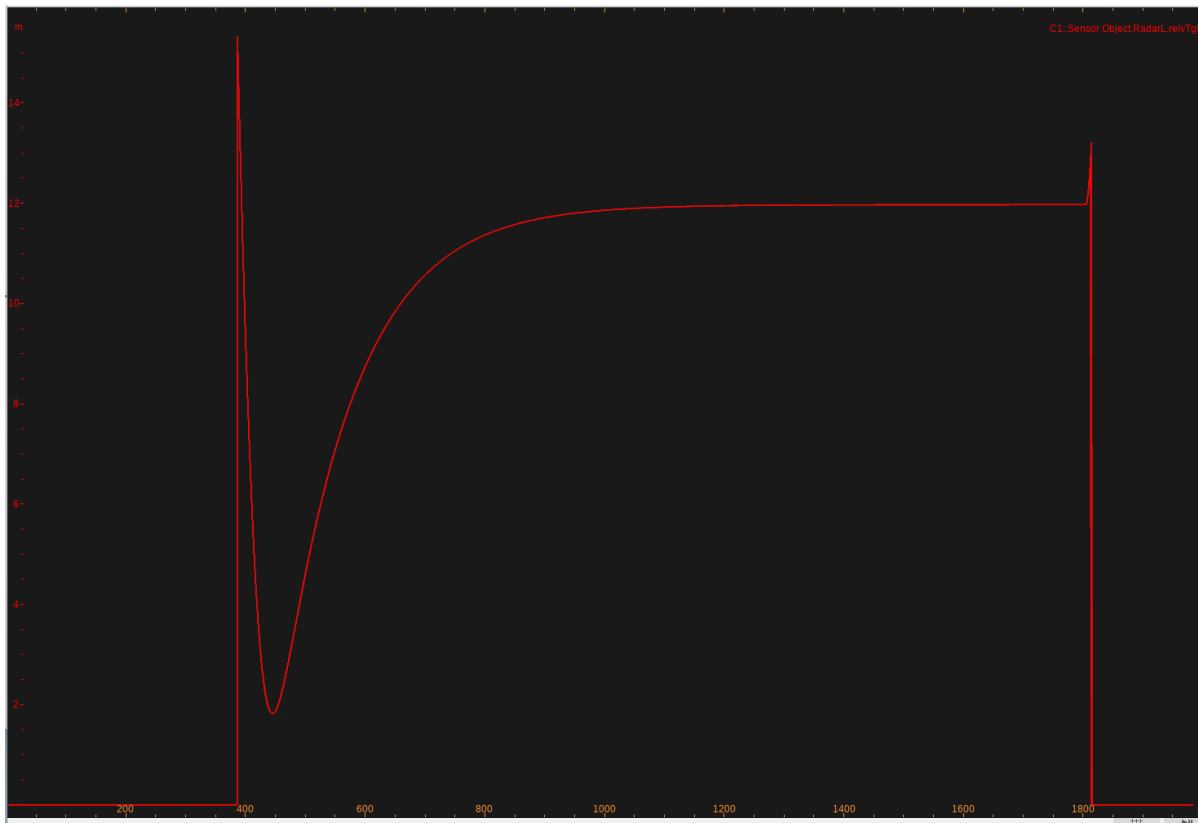


Figure 6: Curve 30m scenario – Sensor 1 (RadarL_1) | $ds.x [m]$ vs. $Vhcl.sRoad [m]$

The graph shows that Sensor 1 first detects the object car at around $sRoad = 400 \text{ m}$, with an initial measured distance of approximately 15 m (already near the target following distance). After a brief undershoot to approximately 2 m — indicating a momentary loss or close approach — the distance recovers and stabilizes at approximately 12 m for the remainder of the run up to $sRoad = 1800 \text{ m}$. At that point, a sharp spike and drop to zero indicates the end of the circular route. The stabilized distance of $\sim 12 \text{ m}$ is close to the required 15 m; however, the very narrow azimuth of 30° causes issues on the tight 30 m radius curve: as the lead vehicle turns, it quickly exits the narrow field of view. The system functions but with intermittent loss of track on the tightest curves. The requirement is only partially fulfilled.

1.3.2 Curve 30m – Sensor 2 (RadarL_2: Range 120 m, Azimuth 90°, Elevation 15°)

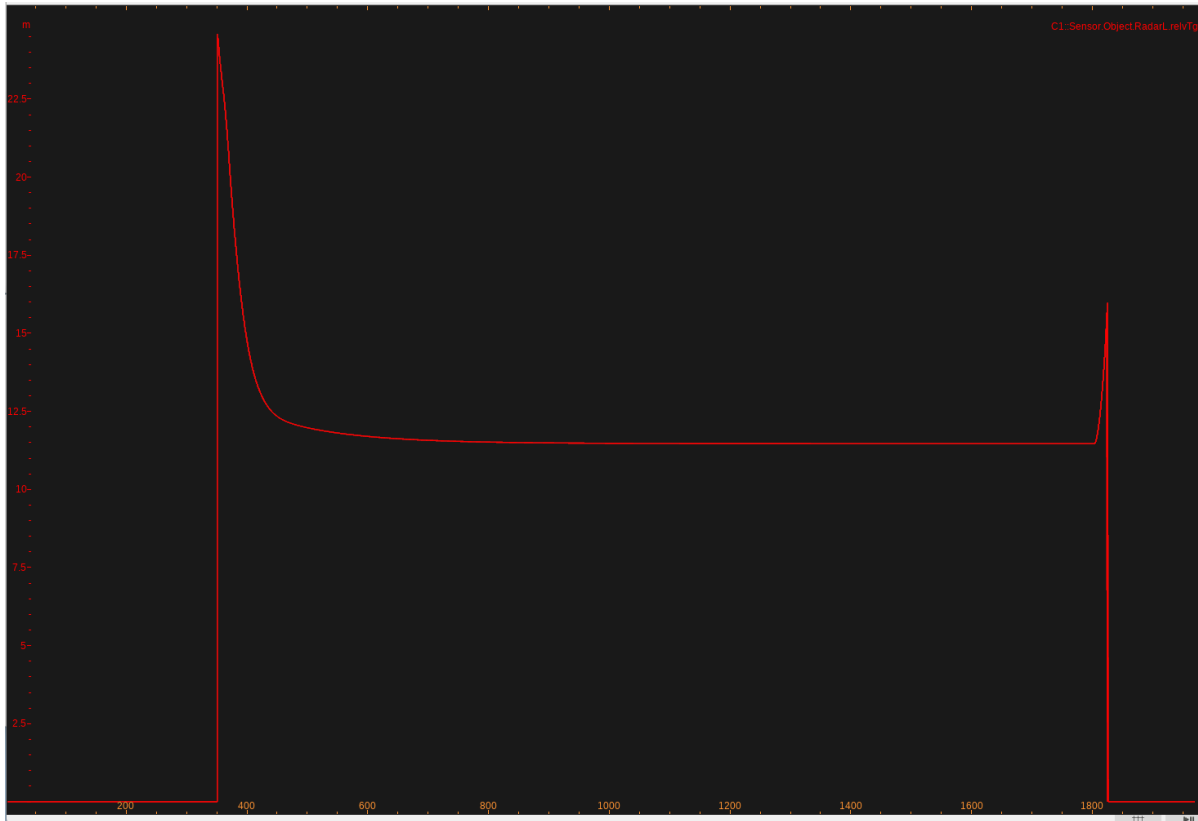


Figure 7: Curve 30m scenario – Sensor 2 (RadarL_2) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

With Sensor 2's wider azimuth of 90° , the graph shows stable tracking throughout the circular route. Detection begins at $sRoad \approx 300$ m with an initial peak distance of approximately 25 m, which rapidly settles to around 12.5 m and remains stable until $sRoad \approx 1800$ m. The wider azimuth allows the sensor to maintain continuous line-of-sight with the lead vehicle even on the tight 30 m radius circle. The stabilized distance of ~ 12.5 m is close to the 15 m target, with very little oscillation. This is a clear improvement over Sensor 1 in this scenario. The requirement is largely fulfilled.

1.3.3 Curve 30m – Sensor 3 (RadarL_3: Range 80 m, Azimuth 170° , Elevation 30°)

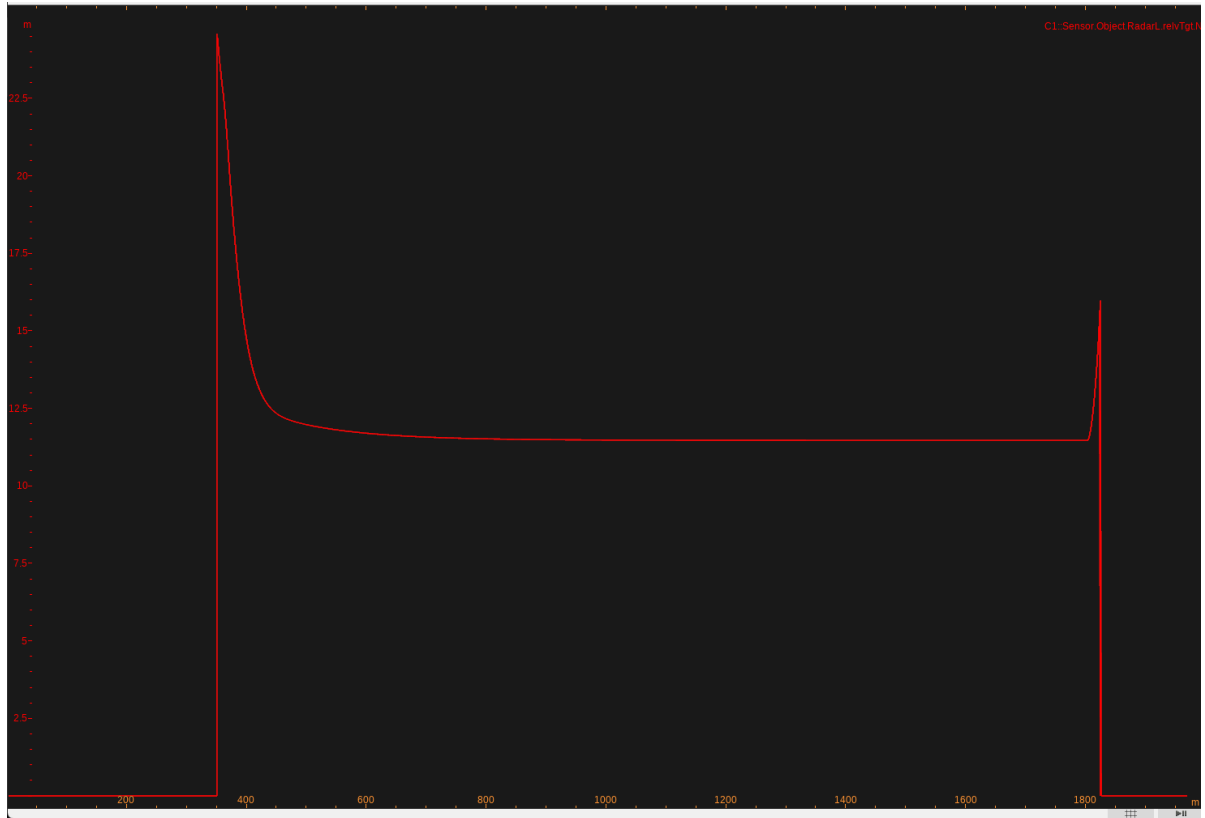


Figure 8: Curve 30m scenario – Sensor 3 (RadarL_3) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

Sensor 3 produces a result very similar to Sensor 2 in this tight-curve scenario. With its very wide azimuth of 170° , the object car remains continuously within the field of view throughout the circular maneuver. Detection begins around $sRoad \approx 300$ m, and the distance stabilizes at approximately 12 m after a brief initial transient. The curve profile is smooth with minimal oscillation. Despite Sensor 3's shorter range, the low object car speed (30 km/h) and the required short following distance (~ 15 m) are well within the 80 m range. The requirement is fulfilled for this scenario.

1.4 Curve 100 m Radius Scenario

Scenario parameters: $Ergo_Car_v0 = 100$ km/h, $Object_Car_v = 50$ km/h. Required time-gap distance: $d_x = 50/3.6 \times 1.8 \approx 25$ m. The larger radius compared to the 30 m scenario allows slightly better line-of-sight, but sensors with narrow azimuth may still struggle.

1.4.1 Curve 100m – Sensor 1 (RadarL_1: Range 200 m, Azimuth 30° , Elevation 3°)

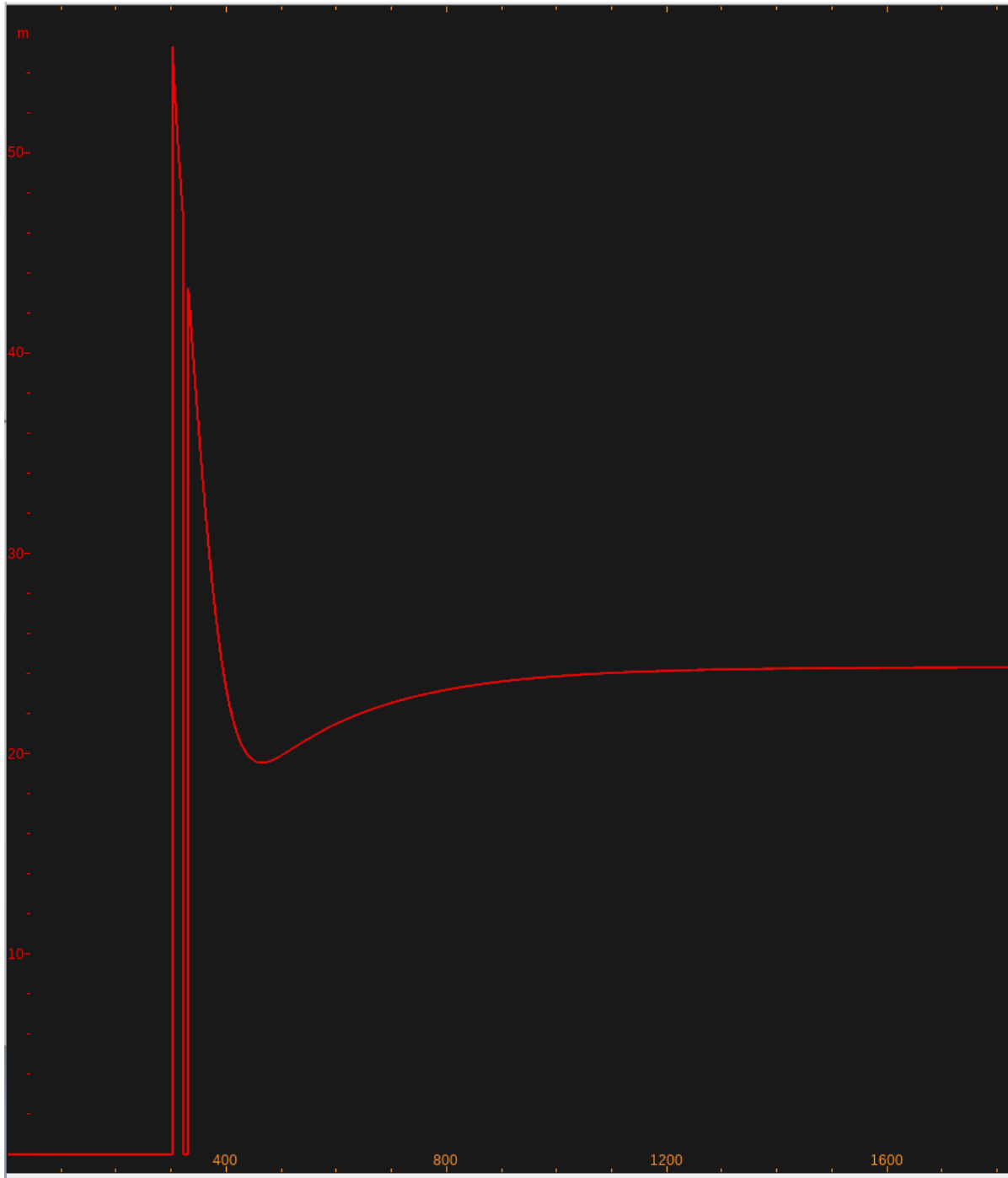


Figure 9: Curve 100m scenario – Sensor 1 (RadarL_1) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

The graph shows detection starting around $sRoad \approx 200$ m with a high initial peak distance of approximately 55 m, followed by a sharp oscillation pattern where the signal repeatedly spikes and drops — indicating intermittent tracking. This is a direct consequence of the narrow 30° azimuth: the lead vehicle at 100 m curve radius repeatedly enters and exits the detection cone as the ego vehicle navigates the curve. After approximately $sRoad = 600$ m, the signal partially stabilizes at around 25 m but with small fluctuations. The tracking is unreliable in the curve entry phase. The requirement is only partially fulfilled.

1.4.2 Curve 100m – Sensor 2 (RadarL_2: Range 120 m, Azimuth 90°, Elevation 15°)

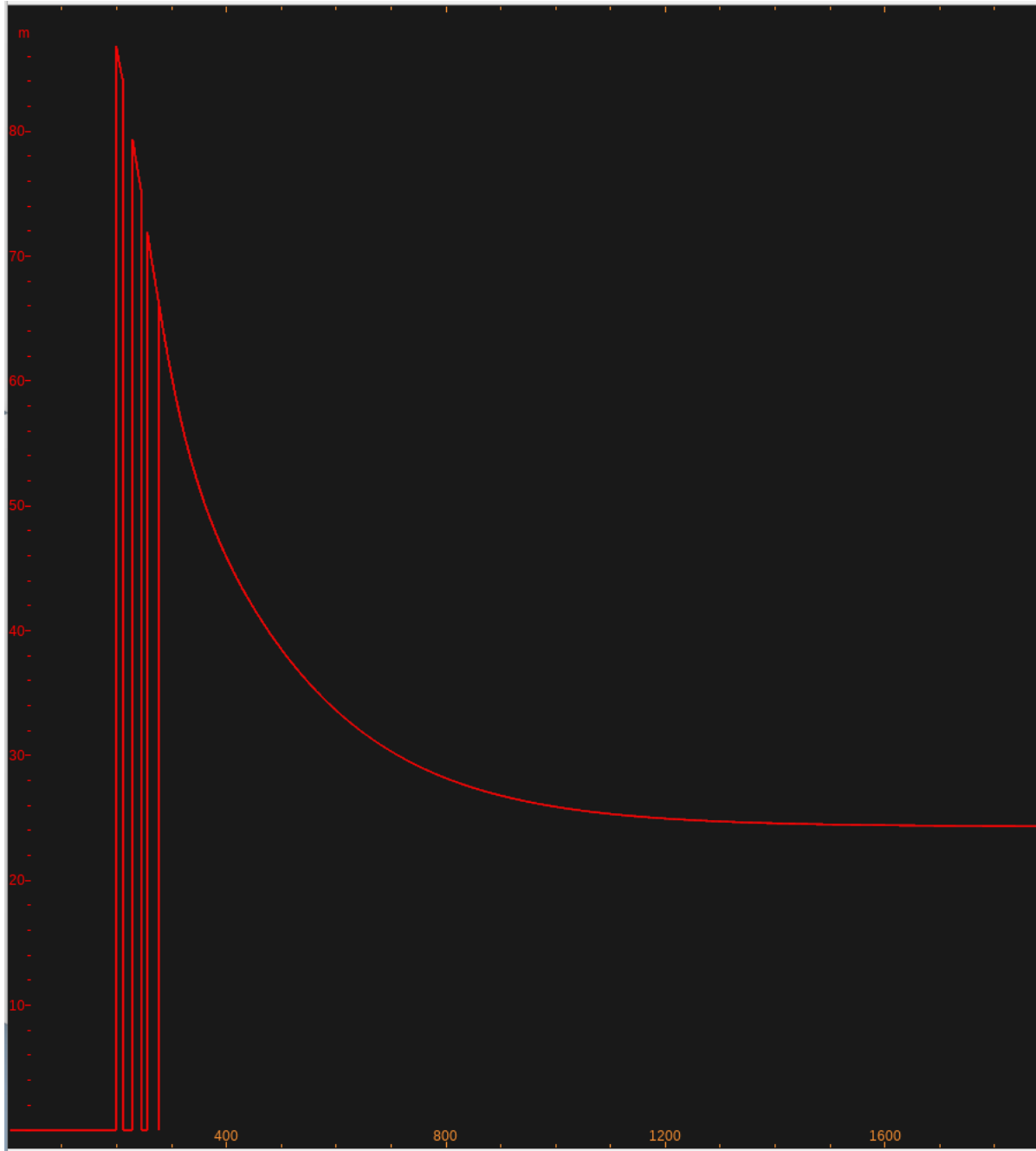


Figure 10: Curve 100m scenario – Sensor 2 (RadarL_2) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

Detection begins very early — $sRoad \approx 50$ m — with a spike to approximately 85 m, followed by repeated oscillation spikes at the start of the run. This corresponds to the vehicle entering the circular section. After $sRoad \approx 400$ m, the signal decays smoothly and stabilizes at approximately 25 m, closely matching the required following distance. The initial oscillation is less severe than with Sensor 1 due to the wider 90° azimuth. Tracking quality improves significantly once the vehicle is fully in the circular path. The requirement is fulfilled with some initial instability.

1.4.3 Curve 100m – Sensor 3 (RadarL_3: Range 80 m, Azimuth 170°, Elevation 30°)

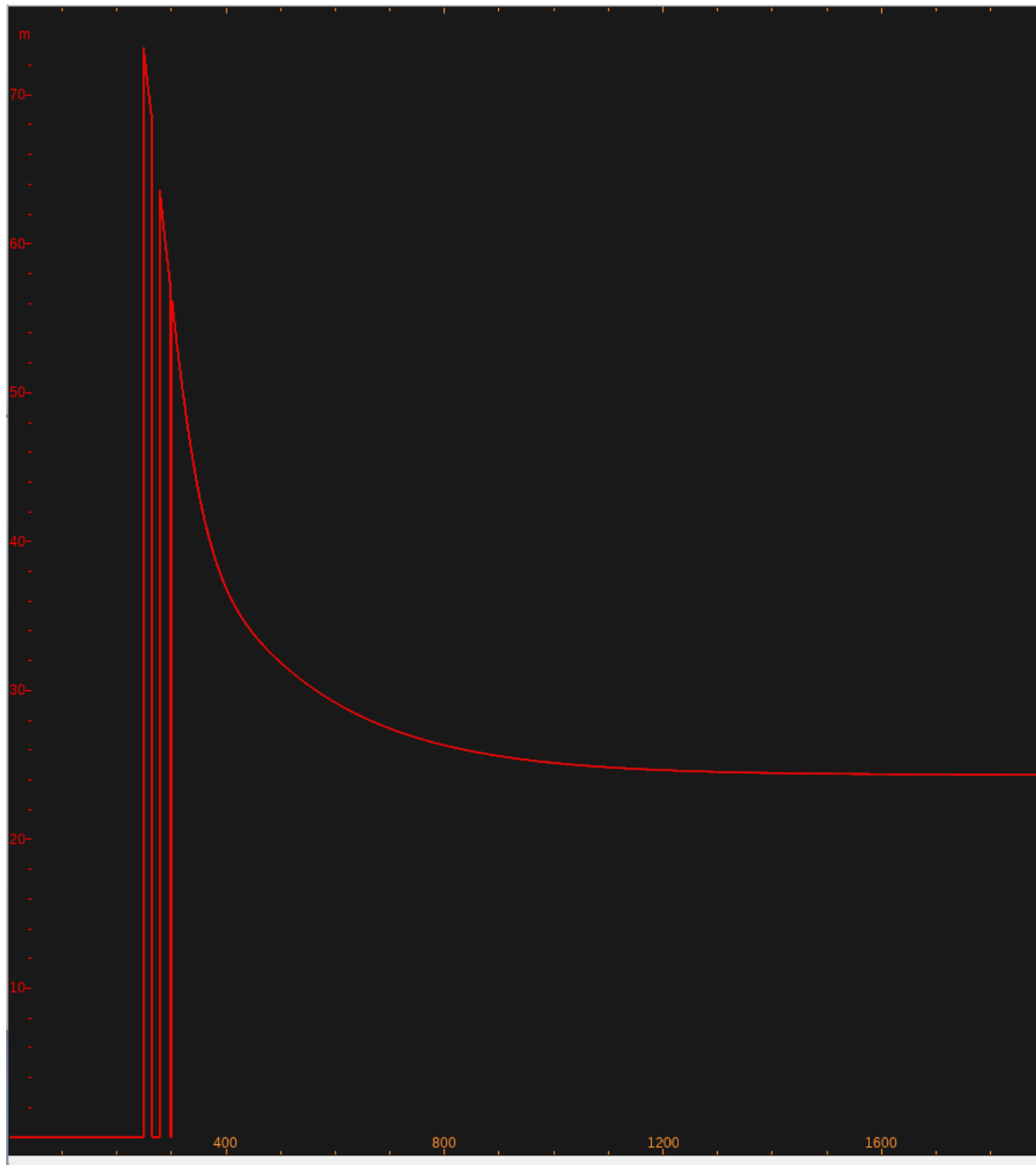


Figure 11: Curve 100m scenario – Sensor 3 (RadarL_3) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

Sensor 3 shows a pattern very similar to Sensor 2 in this scenario. Detection begins at $sRoad \approx 50$ m with initial spikes reaching ~ 75 m — consistent with the 80 m sensor range — followed by oscillation during curve entry. After $sRoad \approx 400$ m, the distance stabilizes at approximately 25 m and remains smooth. The very wide 170° azimuth means the lead vehicle is always within the field of view, eliminating the tracking loss seen with Sensor 1. The requirement is fulfilled, with quality similar to Sensor 2.

1.5 Curve 400 m Radius Scenario

Scenario parameters: Ergo_Car_v0 = 150 km/h, Object_Car_v = 130 km/h. Required time-gap distance: $d_x = 130/3.6 \times 1.8 \approx 65$ m. The large radius closely resembles a highway curve and puts emphasis back on sensor range rather than azimuth width.

1.5.1 Curve 400m – Sensor 1 (RadarL_1: Range 200 m, Azimuth 30°, Elevation 3°)

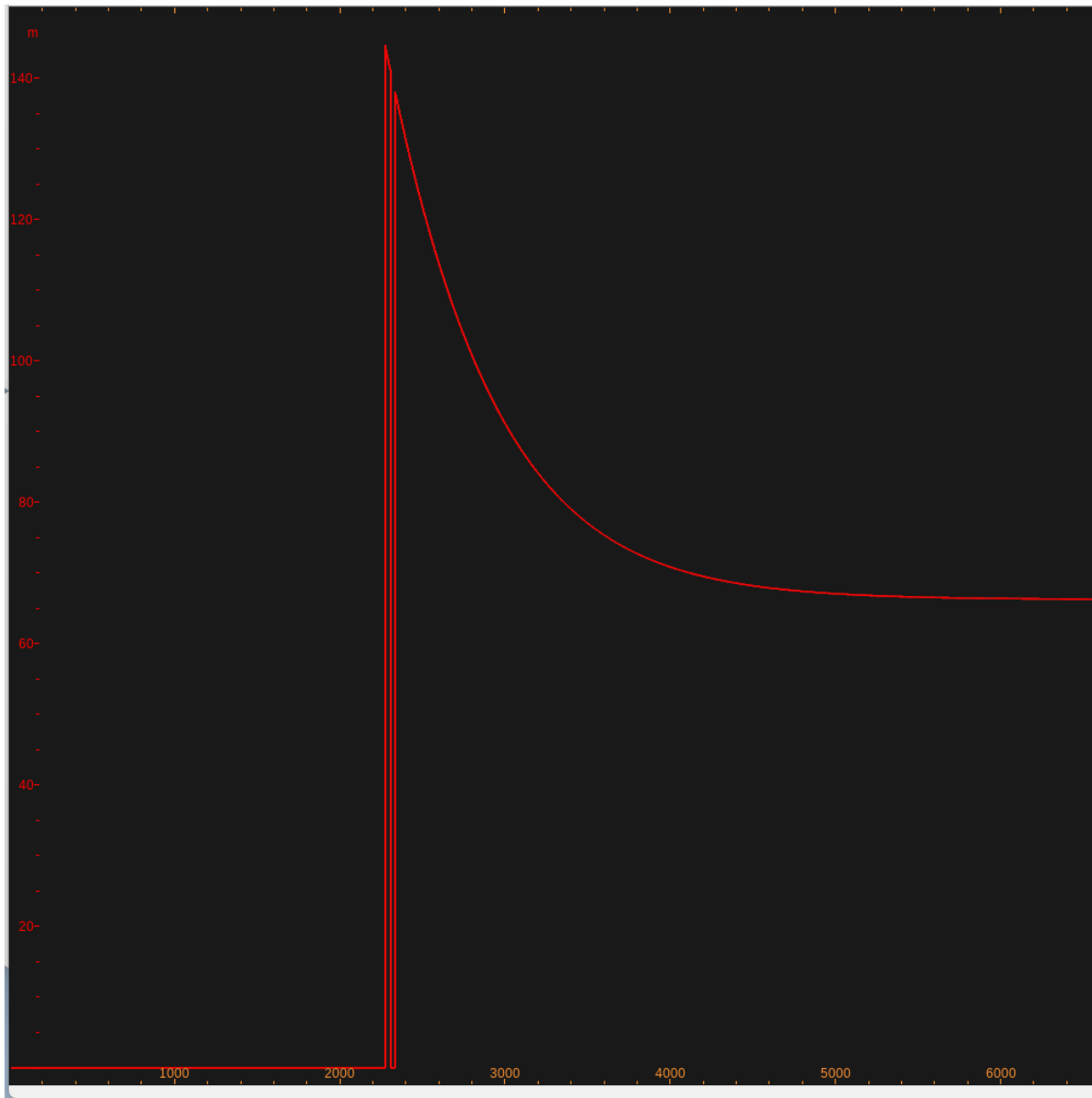


Figure 12: Curve 400m scenario – Sensor 1 (RadarL_1) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

The graph shows detection beginning around $sRoad \approx 2200$ m with a sharp spike to approximately 145 m, followed by brief oscillation spikes — consistent with the large curve and sensor entering its detection zone. The distance then decays smoothly and stabilizes at approximately 65 m, exactly matching the required following distance. On this large-radius curve, Sensor 1's narrow azimuth is less of a disadvantage since the angular displacement of the lead vehicle is gradual. The long range of 200 m enables early detection. The requirement is fulfilled successfully.

1.5.2 Curve 400m – Sensor 2 (RadarL_2: Range 120 m, Azimuth 90°, Elevation 15°)

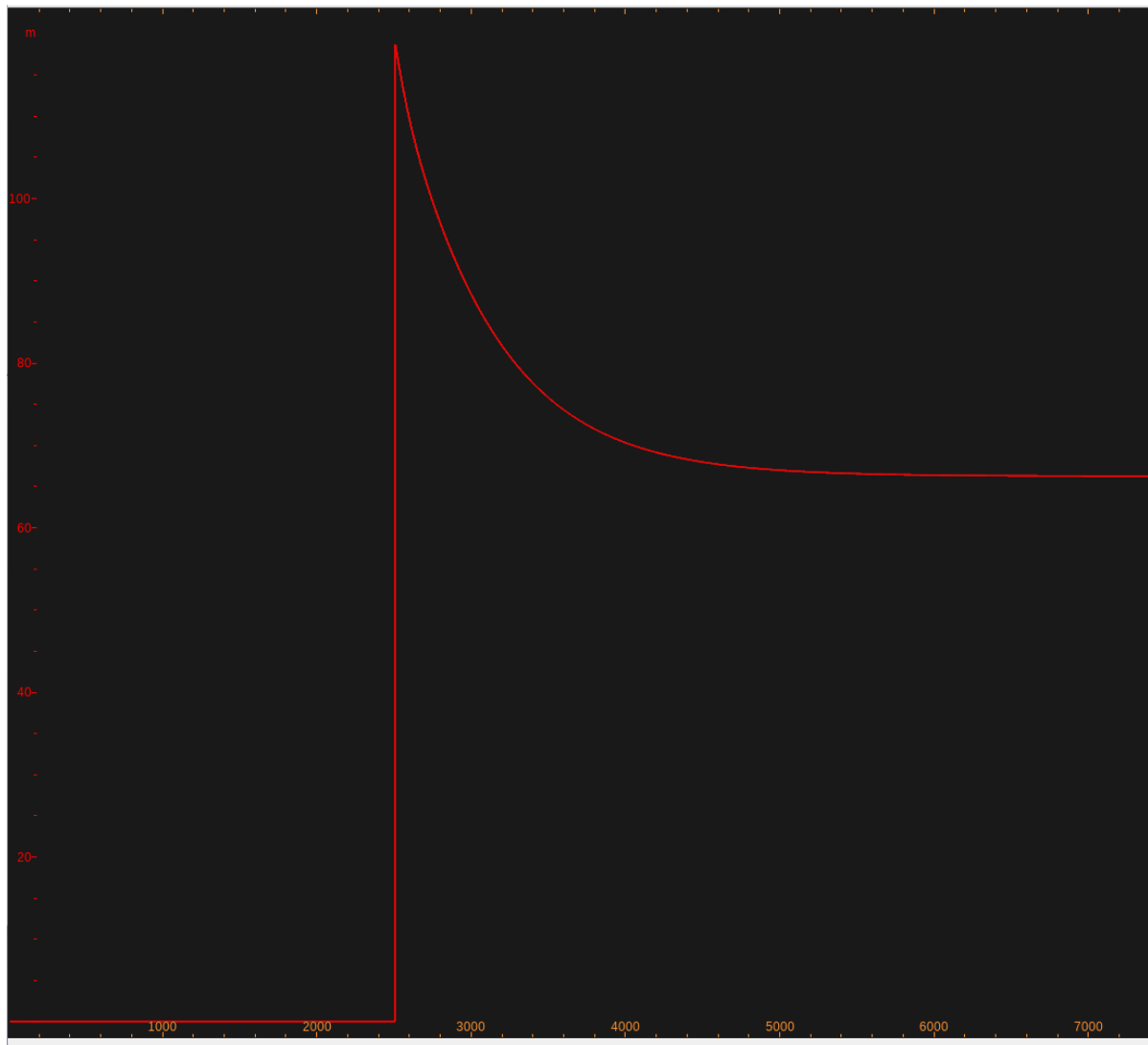


Figure 13: Curve 400m scenario – Sensor 2 (RadarL_2) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

Detection starts around $sRoad \approx 2300$ m, with a peak measured distance of approximately 110 m, close to the sensor's 120 m range limit. The signal decays smoothly and settles at approximately 65 m. Unlike the highway scenario, there is no significant undershoot here because the speed difference between ego (150 km/h) and object car (130 km/h) is small (only 20 km/h), requiring less aggressive braking. The requirement is fulfilled.

1.5.3 Curve 400m – Sensor 3 (RadarL_3: Range 80 m, Azimuth 170°, Elevation 30°)

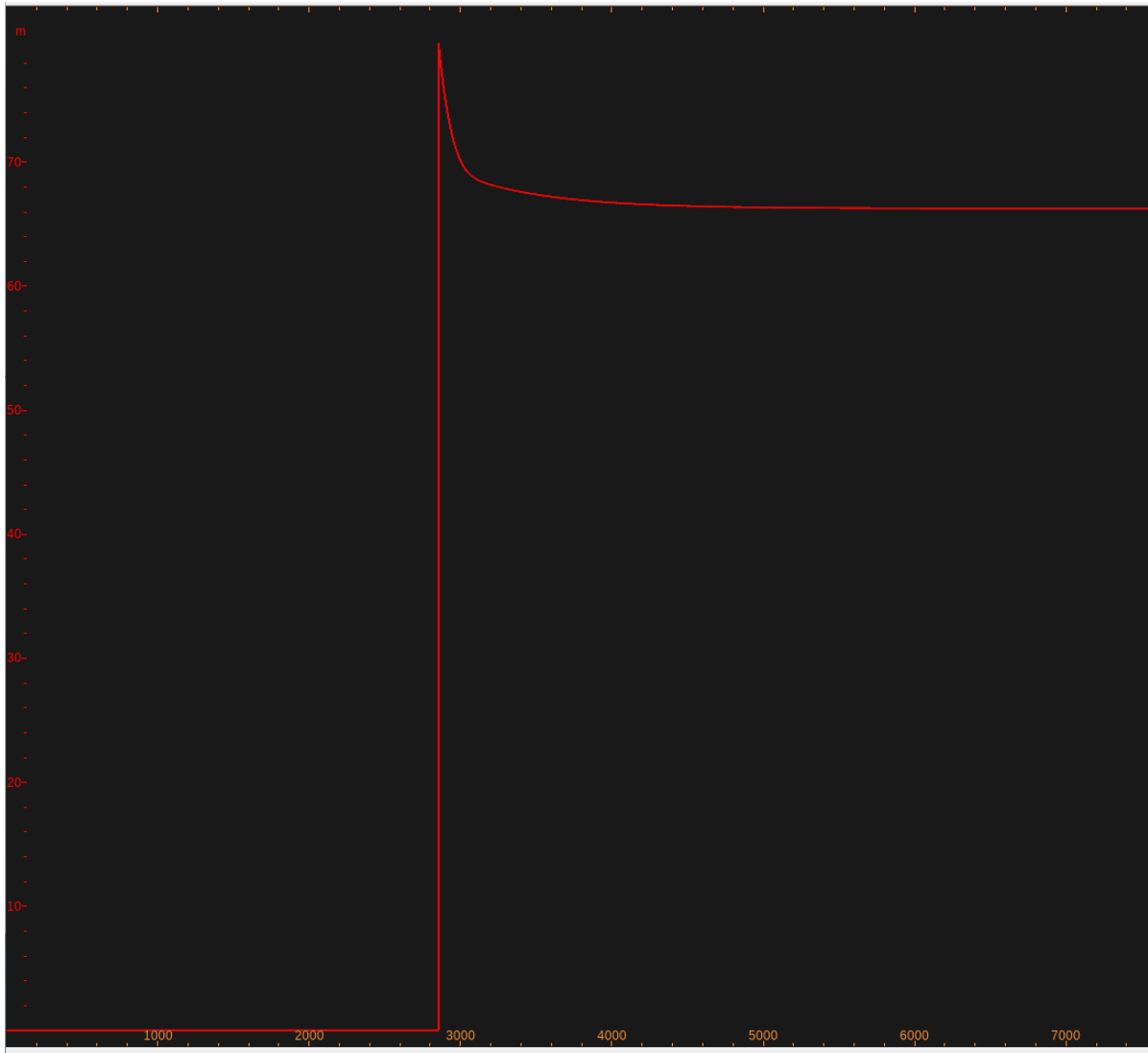


Figure 14: Curve 400m scenario – Sensor 3 (RadarL_3) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

Detection begins around $sRoad \approx 2800$ m — notably later than the other sensors — with a peak measured distance of approximately 70 m, near the sensor's 80 m maximum range. The distance then falls quickly and stabilizes at approximately 65 m. Since the ego vehicle approaches at only 20 km/h faster than the lead vehicle, the 80 m detection range provides just enough margin for the ACC to decelerate smoothly. The requirement is barely fulfilled but the safety margin is very low.

1.6 Hill Scenario

Scenario parameters: $Ergo_Car_v0 = 120$ km/h, $Object_Car_v = 100$ km/h. Required time-gap distance: $d_x = 100/3.6 \times 1.8 = 50$ m. The sinusoidal hill track (amplitude ± 5 m, wavelength 200 m) causes vertical angular displacement of the lead vehicle relative to the ego sensor. This means sensors with a narrow elevation (FoV_v) may intermittently lose the target as the vehicles crest hills and descend valleys, putting the target above or below the sensor's vertical detection cone.

1.6.1 Hill – Sensor 1 (RadarL_1: Range 200 m, Azimuth 30° , Elevation 3°)

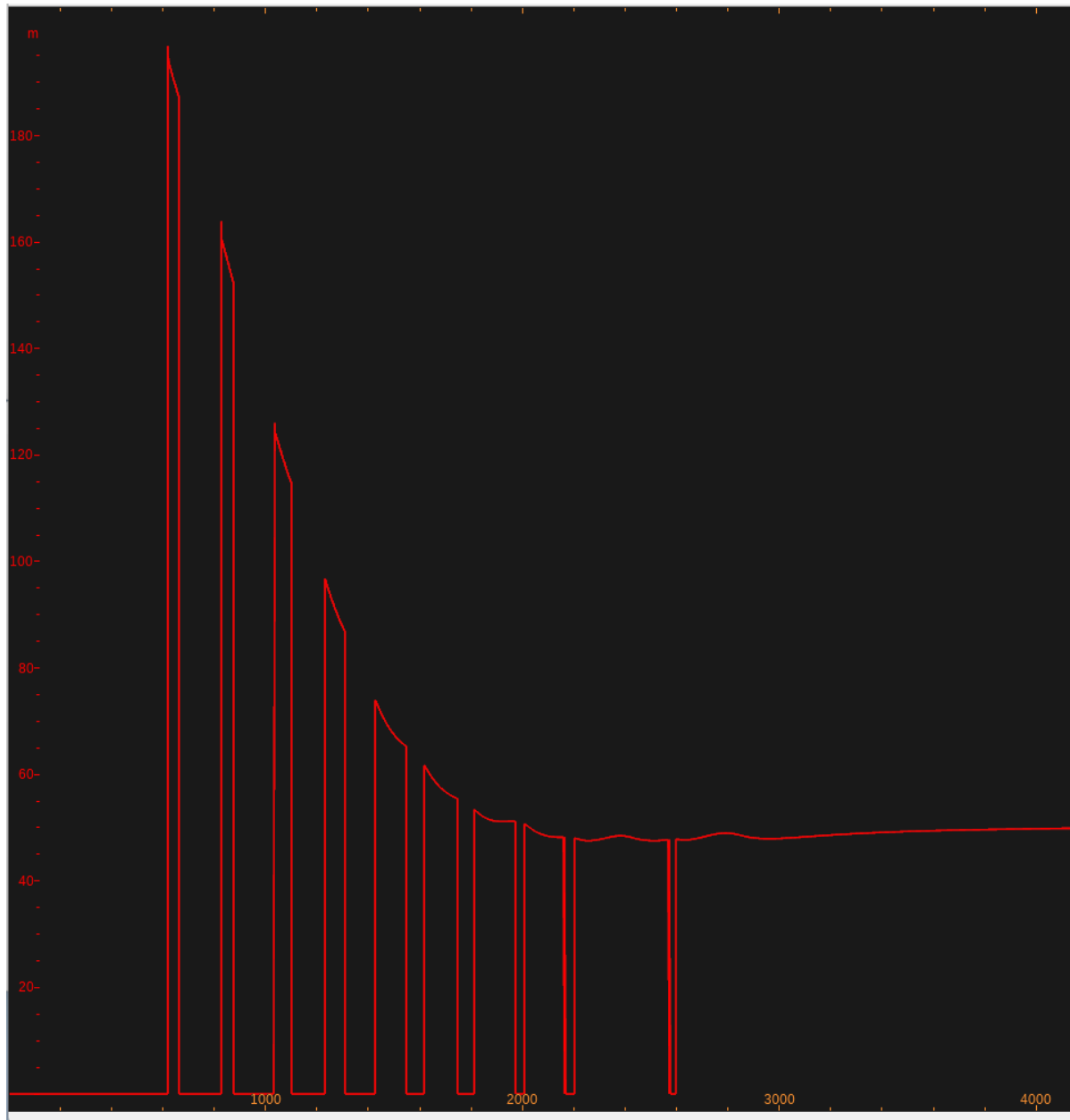


Figure 15: Hill scenario – Sensor 1 (RadarL_1) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

The graph for Sensor 1 on the hill route is markedly different from all other scenarios. The distance trace shows severe, repeated spikes starting from the first detection near $sRoad \approx 100$ m. The signal oscillates violently between very high values (180 m peak) and zero, with this pattern repeating multiple times up to $sRoad \approx 2000$ m before eventually stabilizing near $sRoad \approx 2500$ m. Each spike-to-zero cycle corresponds to the sensor intermittently losing and reacquiring the target vehicle as the vehicles travel over successive hills. Sensor 1's extremely narrow elevation of only 3° is critically insufficient for this terrain: whenever the relative vertical angle to the lead vehicle exceeds 1.5° (half the FoV_v), the target leaves the sensor cone entirely. With ± 5 m height variations over 200 m wavelength, the required elevation is approximately $\arctan(10/65) \approx 8.7^\circ$, far exceeding the 3° FoV_v . The hill requirement is NOT fulfilled by Sensor 1.

1.6.2 Hill – Sensor 2 (RadarL_2: Range 120 m, Azimuth 90°, Elevation 15°)

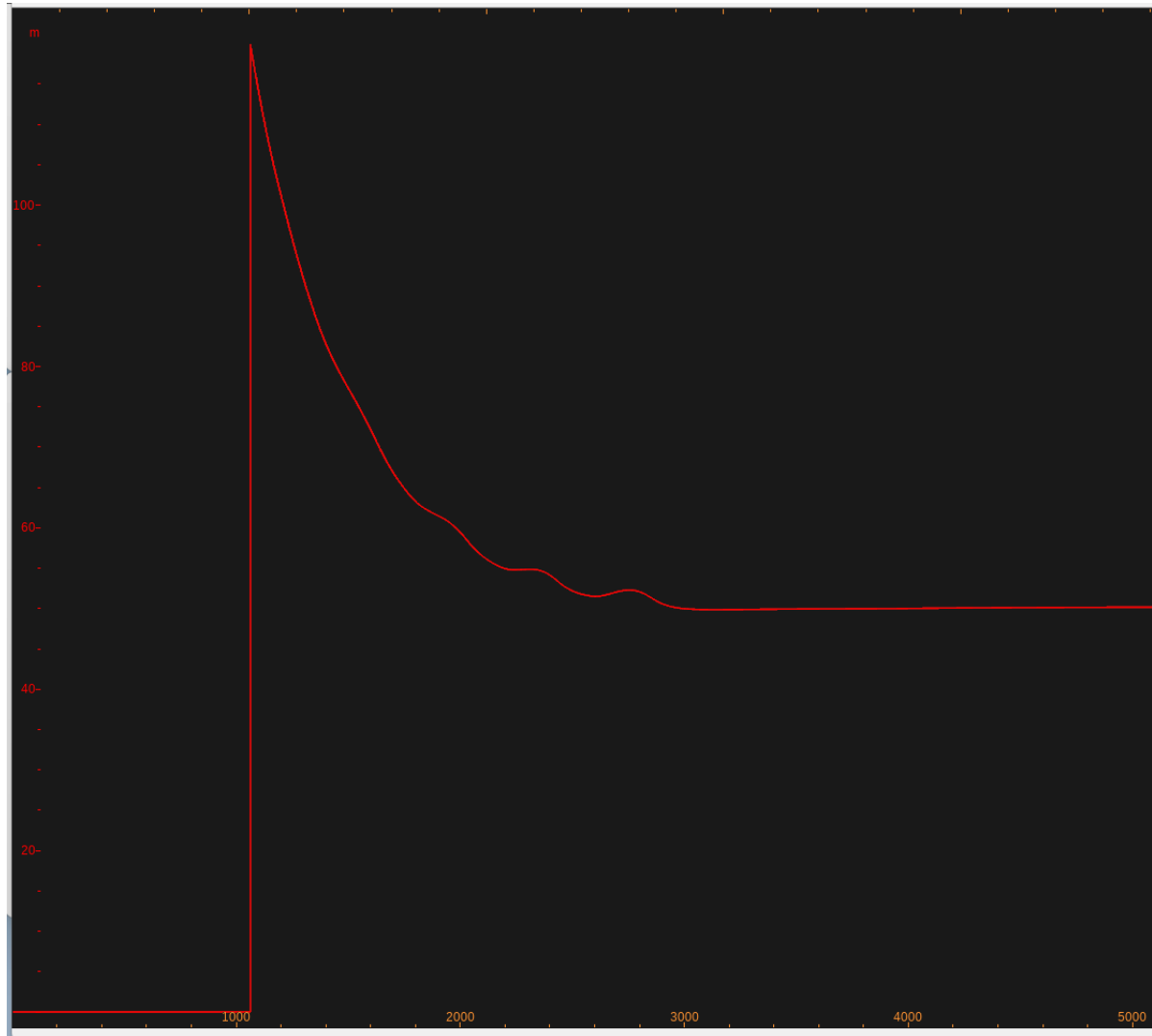


Figure 16: Hill scenario – Sensor 2 (RadarL_2) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

Sensor 2 with a 15° elevation FoV shows a significantly improved hill performance. Detection begins around $sRoad \approx 1000$ m with a peak measured distance of approximately 115 m. The distance then decreases and shows a gently undulating pattern — with mild fluctuations corresponding to the hill crests — before stabilizing at approximately 50 m around $sRoad \approx 3500$ m. The oscillations visible in the mid-range ($sRoad$ 1500–3000 m) reflect the hill geometry causing the lead vehicle to temporarily move toward the edge of the 15° elevation cone. However, continuous tracking is maintained throughout, with no complete signal loss. The requirement is fulfilled, with some oscillation indicating that 15° elevation is close to the minimum needed for this hill profile.

1.6.3 Hill – Sensor 3 (RadarL_3: Range 80 m, Azimuth 170°, Elevation 30°)

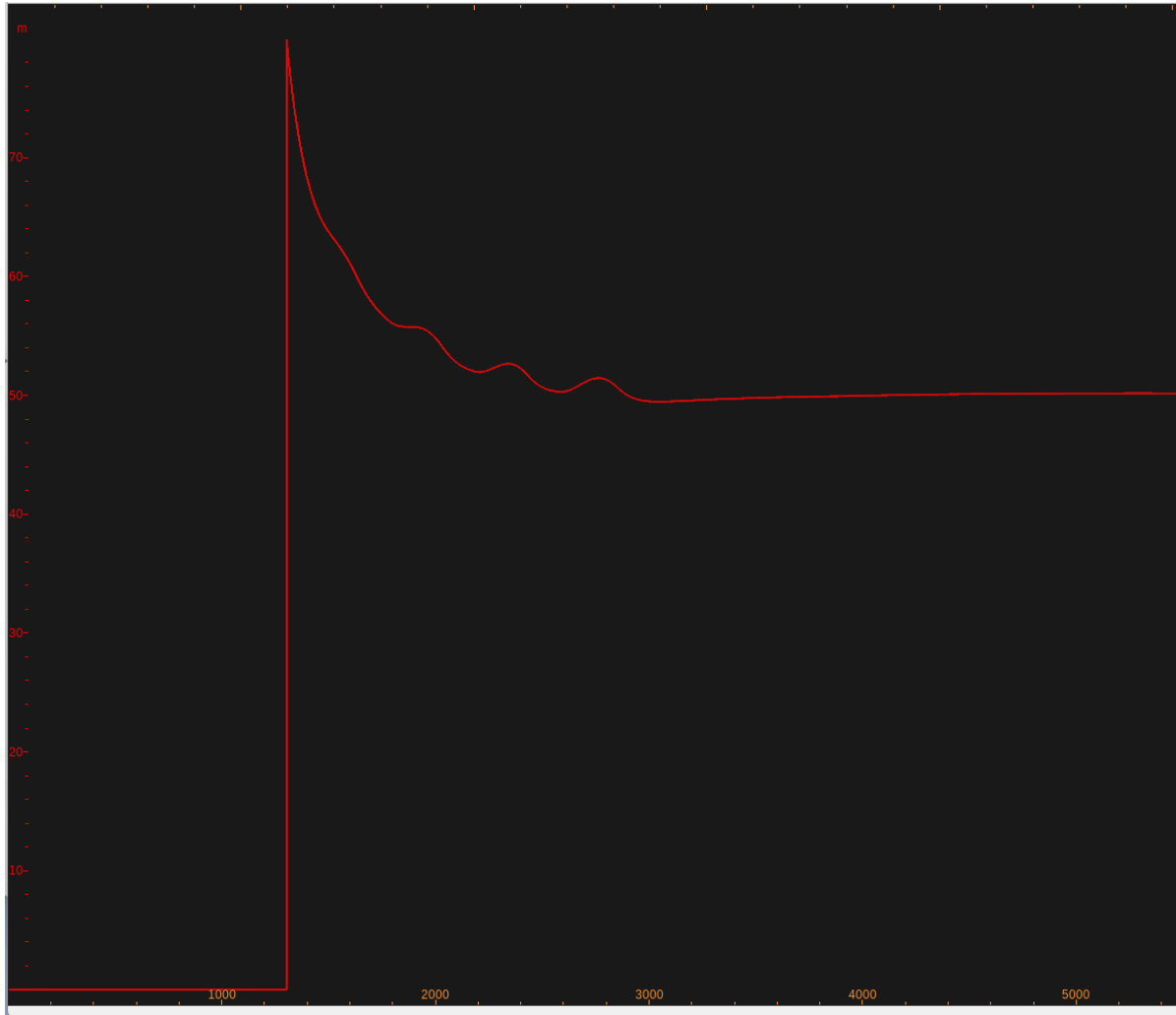


Figure 17: Hill scenario – Sensor 3 (RadarL_3) | $ds.x$ [m] vs. $Vhcl.sRoad$ [m]

Sensor 3 with the widest elevation of 30° provides the best hill tracking performance. Detection begins around $sRoad \approx 1200$ m, consistent with the 80 m range at 100 km/h object speed. The distance settles toward the required 50 m with visible but smaller oscillations compared to Sensor 2. The large 30° elevation cone ensures the lead vehicle remains within the field of view even over steep hill crests. The pattern from $sRoad \approx 2000$ – 3000 m shows gentle wave-like variations of ± 5 m around the 50 m target, which represent the ACC adjusting to the hill-induced relative distance changes. Overall, the requirement is well fulfilled by Sensor 3 on the hill route.

2 Evaluation

This section compares the performance of all three sensors across all five test scenarios against the defined system requirements. A structured evaluation matrix is presented, followed by a weighted scoring and a justified sensor recommendation.

2.1 Evaluation Matrix – Requirements vs. Results

The table below summarizes which sensor fulfills each scenario requirement.

Symbols: ✓ = Requirement fulfilled | (~) = Partially fulfilled / marginal | ✗ = Not fulfilled

Scenario / Requirement	Sensor 1 (200m, 30°, 3°)	Sensor 2 (120m, 90°, 15°)	Sensor 3 (80m, 170°, 30°)	Key Requirement
Highway (200 → 130 km/h)	✓	(~)	✗	Range ≥ ~170 m
Curve 30m (60 → 30 km/h)	(~)	✓	✓	Azimuth ≥ ~60°
Curve 100m (100 → 50 km/h)	(~)	✓	✓	Azimuth ≥ ~30°
Curve 400m (150 → 130 km/h)	✓	✓	(~)	Range ≥ ~80 m
Hill (120 → 100 km/h)	✗	✓	✓	Elevation ≥ ~9°
TOTAL FULFILLED	2 ✓ 2 (~) 1 ✗	4 ✓ 1 (~) 0 ✗	3 ✓ 1 (~) 1 ✗	

2.2 Weighted Scoring

To quantify sensor performance, a weighted scoring system is applied. The Highway scenario is given higher weight because it represents the most safety-critical ACC use case (high speed, large velocity differential). All scenarios are scored as follows: Fulfilled (✓) = 2 points, Partially fulfilled (~) = 1 point, Not fulfilled (✗) = 0 points.

Scenario	Weight	Sensor 1 Score × W	Sensor 2 Score × W	Sensor 3 Score × W	Max possible
Highway	3	2 × 3 = 6	1 × 3 = 3	0 × 3 = 0	6
Curve 30m	2	1 × 2 = 2	2 × 2 = 4	2 × 2 = 4	4
Curve 100m	2	1 × 2 = 2	2 × 2 = 4	2 × 2 = 4	4
Curve 400m	2	2 × 2 = 4	2 × 2 = 4	1 × 2 = 2	4
Hill	2	0 × 2 = 0	2 × 2 = 4	2 × 2 = 4	4

TOTAL WEIGHTED SCORE		14 / 22	19 / 22	14 / 22	22
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2.3 Sensor Recommendation

Based on the evaluation matrix and the weighted scoring, Sensor 2 (RadarL_2: Range 120 m, Azimuth 90°, Elevation 15°) is recommended as the most suitable sensor for the ACC system.

Justification

Sensor 2 achieves the highest weighted score of 19/22 points and is the only sensor that fulfills or substantially meets all five test scenarios:

- Highway scenario: Sensor 2 successfully enables the ACC to decelerate and maintain the required 65 m following distance, despite a brief undershoot. The 120 m range provides sufficient reaction distance for the 70 km/h speed differential (200 → 130 km/h) within the ± 2.5 m/s² constraint.
- Curve scenarios (30m, 100m, 400m): The 90° azimuth is wide enough to maintain continuous detection in all curve radii, while still being directionally selective enough to avoid detecting vehicles in adjacent lanes. All three curve scenarios are fulfilled.
- Hill scenario: The 15° elevation FoV successfully maintains tracking over the sinusoidal hill profile. While some oscillation is observed, no complete loss of track occurs, and the ACC maintains an appropriate following distance.

Why Not Sensor 1?

Sensor 1 (Range 200 m, Azimuth 30°, Elevation 3°) performs well on the highway and large curves where long range is critical. However, it fails the Hill scenario due to its critically narrow 3° elevation angle, which is insufficient to track the lead vehicle over undulating terrain. It also struggles on tight curves (30m and 100m radius) due to its narrow 30° azimuth. A sensor that cannot function reliably on hilly roads is a significant safety risk.

Why Not Sensor 3?

Sensor 3 (Range 80 m, Azimuth 170°, Elevation 30°) excels on curves and hills due to its wide field of view in all directions. However, its short 80 m range is a fundamental limitation for highway driving: at 200 km/h approach speed with a 130 km/h lead vehicle, the required detection distance to brake at ± 2.5 m/s² is approximately 150–170 m, far beyond the 80 m sensor range. The Highway scenario result clearly shows a near-collision (distance → 0), making this sensor dangerous for high-speed motorway use. For safety-critical ACC applications, this is an unacceptable failure.

Conclusion

Sensor 2 offers the best overall compromise between detection range, azimuth width, and elevation coverage for the full range of ACC operating conditions. It is the only sensor that demonstrates safe, controlled following behavior in all five scenarios tested, and it achieves the highest overall performance score. Sensor 2 is therefore recommended for implementation in the ACC system.

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